Transformer u(t) First-Layer Experiment Report

Fixed-initial-condition time-dependent control trained by PMP/KKT optimality gaps

This report focuses on the manuscript's time-dependent control strategy u_theta(t): [0,T] -> [0,u_max] for the fixed initial condition. The state-dependent feedback extension u(t,N) is not included here.

1. Model and Training Objective

We use the population dynamics and cost from the manuscript:

$$\dot{N}_i(t) = \big(r_i - \phi_i u(t) - M_i G(N(t))\big) N_i(t) \tag{1}$$

$$G(N) = \log\left(1 + \frac{1}{m}\sum_{k=1}^{m} N_k\right) \tag{2}$$

$$J(u) = \alpha^\top N(T) + \int_0^T \big(\beta^\top N(t) + \gamma u(t)\big)\, dt \tag{3}$$

Reported parameters: T=10, m=21, u_max=3, alpha=1, beta=0.1, gamma=20, and N_i(0)=10. The control is represented by a small Transformer encoder over normalized time.

Given u_theta(t), we roll out N_theta(t), solve the costate equation backward, and compute the switching function

$$\psi(t) = H_u(N, \lambda, u) = \gamma - \sum_i \phi_i \lambda_i(t) N_i(t) \tag{4}$$

component	role
non-singular KKT loss	enforces u=0 when psi>0 and u=u_max when psi<0
singular loss	near psi=0, matches u_theta(t) to the singular candidate u_sing(N(t))

Here q(t) is a smooth weight that switches between the singular condition near psi(t)=0 and the boundary KKT condition away from psi(t)=0.

2. Learned u(t), N(t), and N(u)

The learned control starts high, transitions to a lower interior/singular-like region, and increases again near the terminal portion.

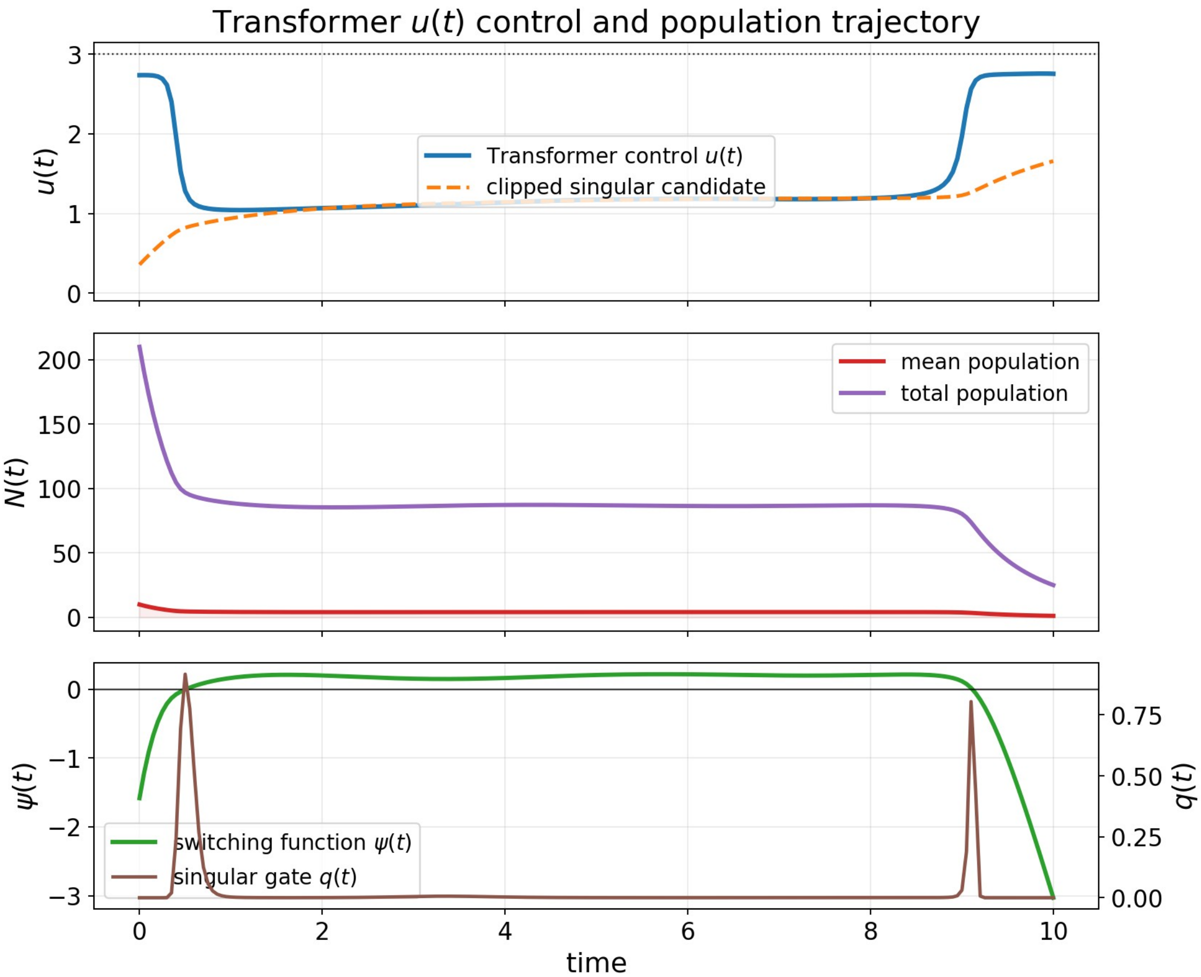

Figure 1. Learned Transformer control u(t), population trajectory N(t), switching function psi(t), and singular weight q(t).

The N(u) phase plot below uses the total population across all 21 subpopulations.

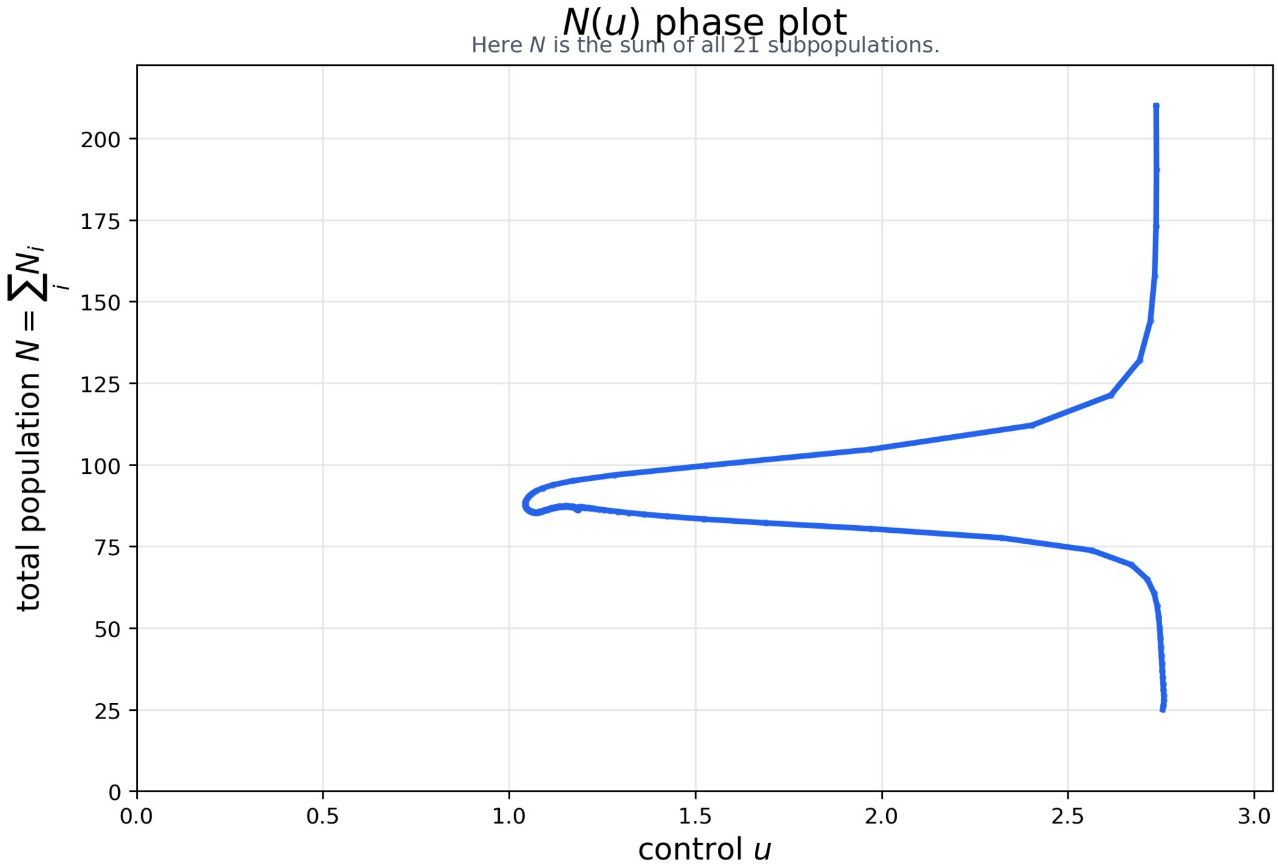

Figure 2. N(u) phase plot using the total population across the 21 subpopulations.

3. Training Loss Trajectories for PMP/KKT Conditions

The table reports the smallest recorded training loss. In this experiment, the training loss is the manuscript's PMP/KKT optimality gap, composed of the singular condition and the non-singular Hamiltonian minimization condition.

metric	value
total training loss / PMP-KKT optimality gap	0.02637
singular-condition training loss	0.00167
non-singular Hamiltonian-minimization training loss	0.02471
objective J on the training grid	384.76
range and mean of u_theta(t)	min 1.038, max 2.758, mean 1.372
terminal mean state	1.207

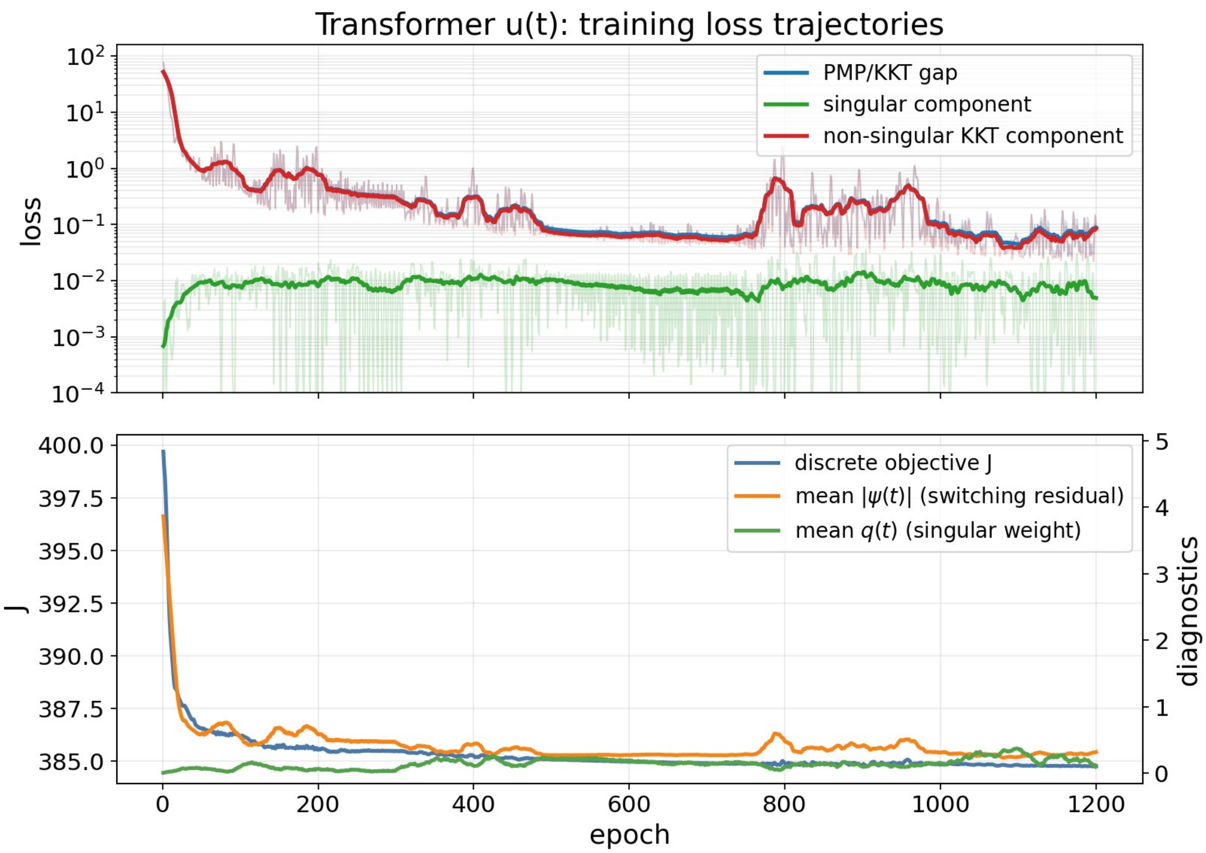

Figure 3. Training trajectories of the total PMP/KKT optimality gap and its singular and non-singular components.

The final pointwise diagnostic below uses a smooth weight q(t) to separate the two regimes: near psi(t)=0 it emphasizes the singular-condition error; away from psi(t)=0 it emphasizes the boundary KKT error.

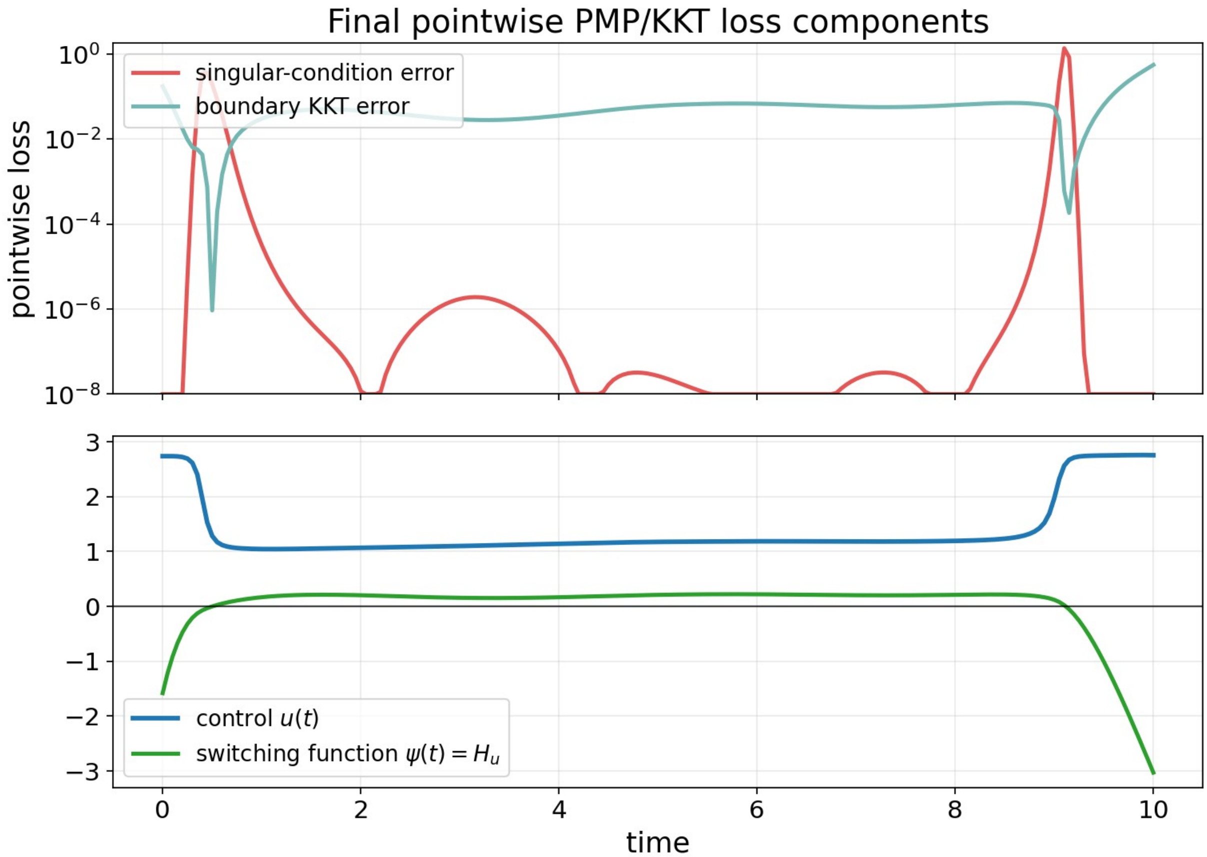

Figure 4. Pointwise singular-condition error and boundary KKT error along the final learned trajectory.

4. First-Layer Benchmark and Related Work

For this first-layer experiment, all numerical comparisons keep the same fixed initial condition and compare time-dependent controls u(t). Several related-work papers in the manuscript target value-function or feedback formulations; those are important method references, but they are not the same numerical task as this u(t) reproduction.

reference	learned object / formulation	relation to this report
HJB / BSDE methods [1,2,7]	value function V(t,N) or HJB PDE solution	state-domain feedback/value formulation; not a direct u(t) baseline
Neural-PMP [3]	control sequence via forward rollout, backward costate recursion, and Hamiltonian-gradient updates	closest related-work baseline for the present u(t) experiment; implemented numerically below
DeepONet / PINN policy iteration [5,6]	policy evaluation and improvement for HJB-type equations	methodological reference for feedback/value learning; separate from this fixed-trajectory u(t) layer
classical chemotherapy OC [4]	PMP and singular-control structure	source of the singular-control condition used in the manuscript

Among these, [3] is the closest apples-to-apples numerical comparison. Here [3] refers to Gu, Xiong, and Chen, Pontryagin Optimal Control via Neural Networks (arXiv:2212.14566). Their Neural-PMP method first learns a differentiable dynamics model and then updates a control sequence using PMP gradients. Since the dynamics are already known in the manuscript setting, we compare against the oracle-dynamics controller stage: forward state integration, backward costate recursion, and Hamiltonian-gradient updates of a discrete control sequence.

method	training / update criterion	objective J	J - direct	PMP/KKT gap	note
direct grid reference	direct minimization of discretized J	386.438	0.000	1.016	cost reference
Transformer u(t), 6 runs	paper PMP/KKT optimality-gap loss	386.738 +/- 0.045	0.300 +/- 0.045	0.463 +/- 0.148	main reproduction
best Transformer run	same as above	386.695	0.257	0.207	best reported run
Neural-PMP [3]	Hamiltonian-gradient update with known dynamics	386.986	0.548	8.574	related-work baseline
constant u=1.5	fixed control	400.403	13.965	76.556	scale check
repository demo output	provided s.csv	422.670	36.232	433.349	original demo artifact

* Objective J is computed after fixing u(t), reintegrating N(t) with the same fine-step fourth-order Runge-Kutta evaluator, and applying the manuscript objective definition. The direct row is a numerical cost reference, not a neural model.

The Transformer runs are stable across random seeds and remain close to the direct cost reference. Compared with the Neural-PMP controller-stage baseline [3], the Transformer has both lower objective J and a smaller PMP/KKT diagnostic gap in this fixed-initial-condition u(t) experiment.

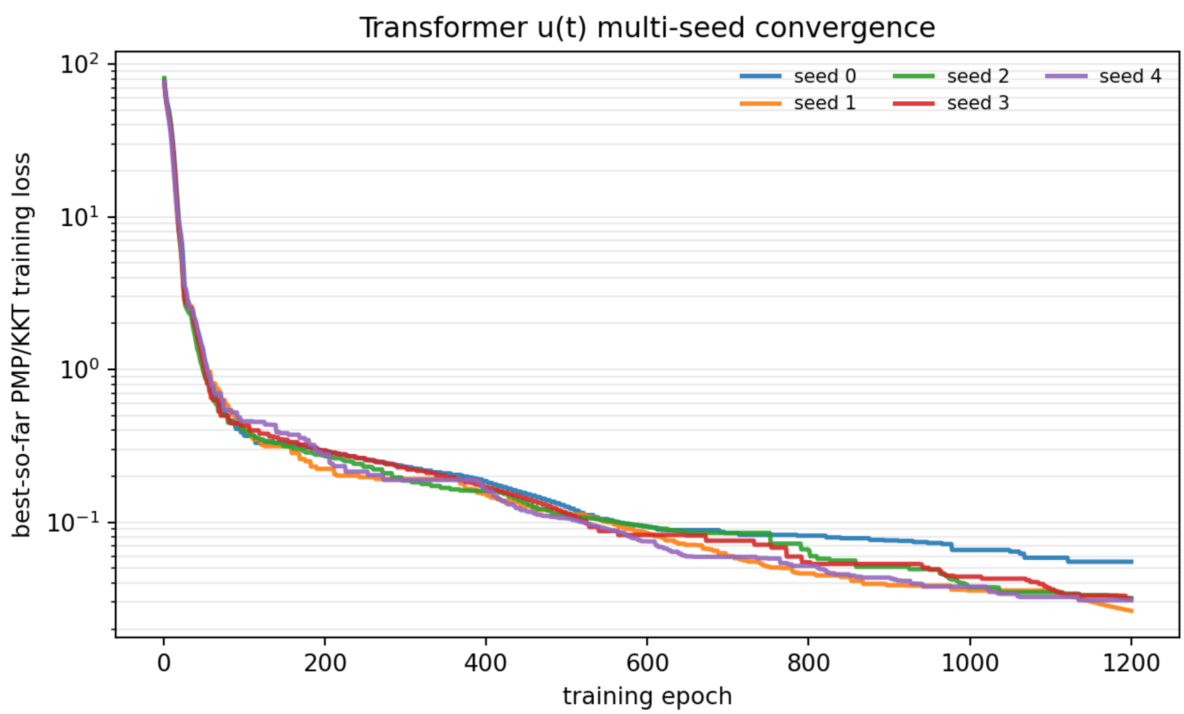

Figure 5. Multi-seed convergence of the Transformer u(t) PMP/KKT training loss.

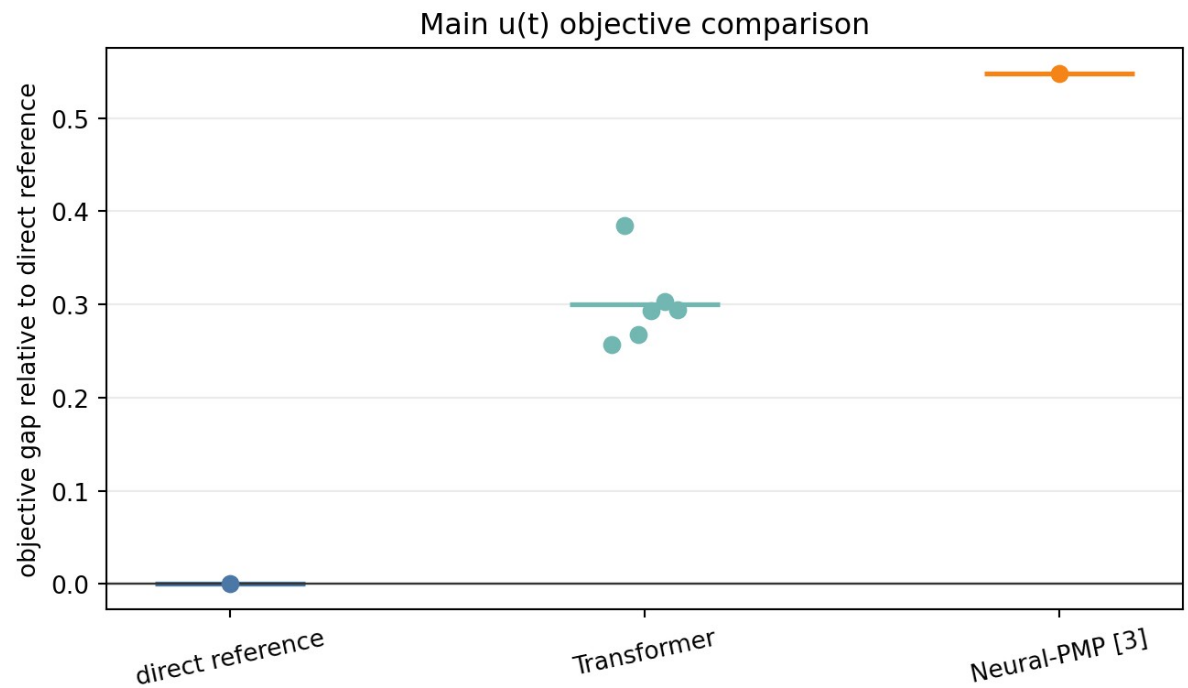

Figure 6. Objective gap relative to the direct cost reference for the main u(t) comparison.

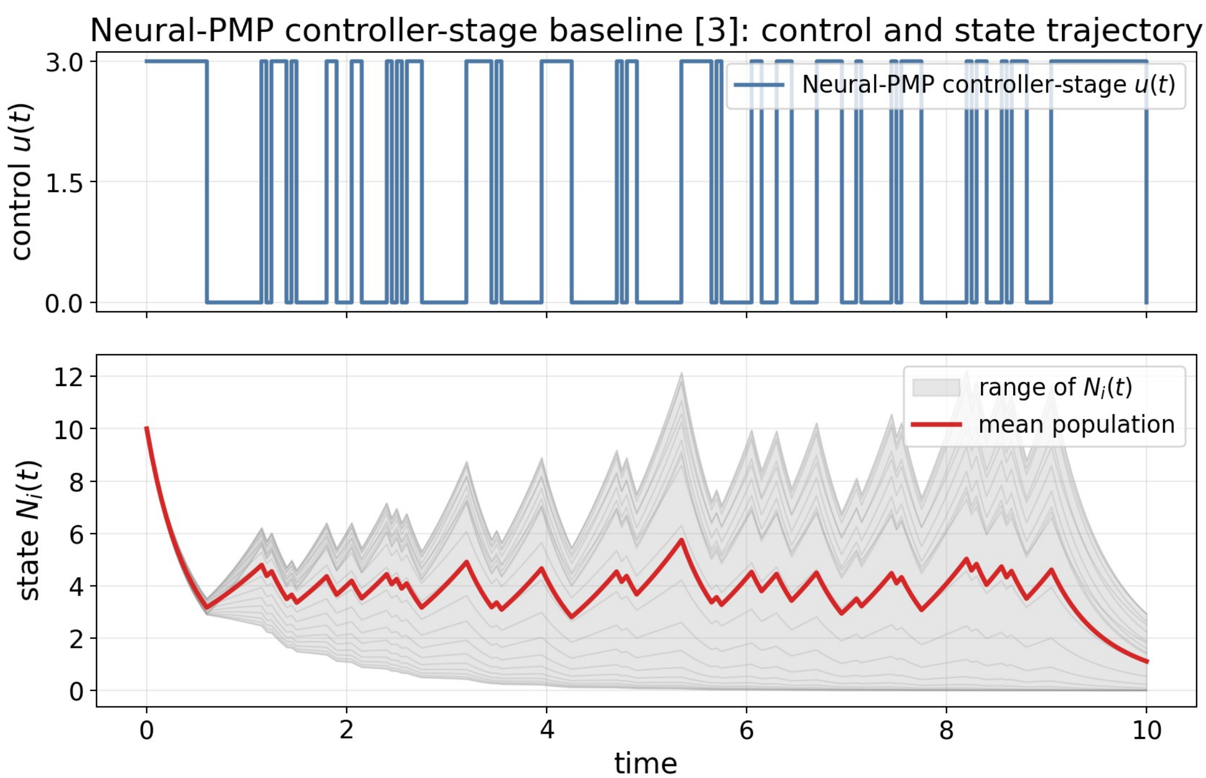

Figure 7. Neural-PMP controller-stage baseline [3]: control sequence and resulting state trajectory.

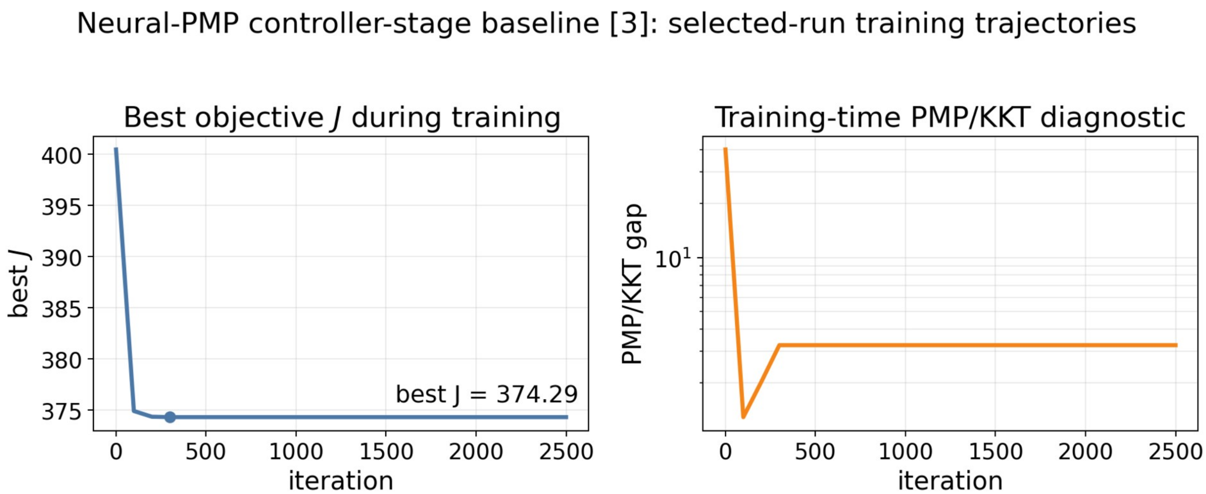

Figure 8. Neural-PMP controller-stage baseline [3]: selected-run training trajectories.

5. Conclusion

The requested first-layer u(t) reproduction is complete. The Transformer control trained with the manuscript's PMP/KKT optimality-gap loss is smooth, satisfies the control bounds, and reduces the training optimality gap from about 76 to 0.026 in the best run. Across six Transformer runs, the common-evaluator objective is 386.738 +/- 0.045, close to the direct cost reference 386.438 and better than the Neural-PMP [3] controller-stage baseline in this setting. The state-dependent extension u(t,N) is left for separate work because it requires different optimality conditions.